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Risa 3D

Analysis of a Stable Frame Structure

AISC-ASD- Code Check

Analysis of a Stable Frame Structure

Procedure:

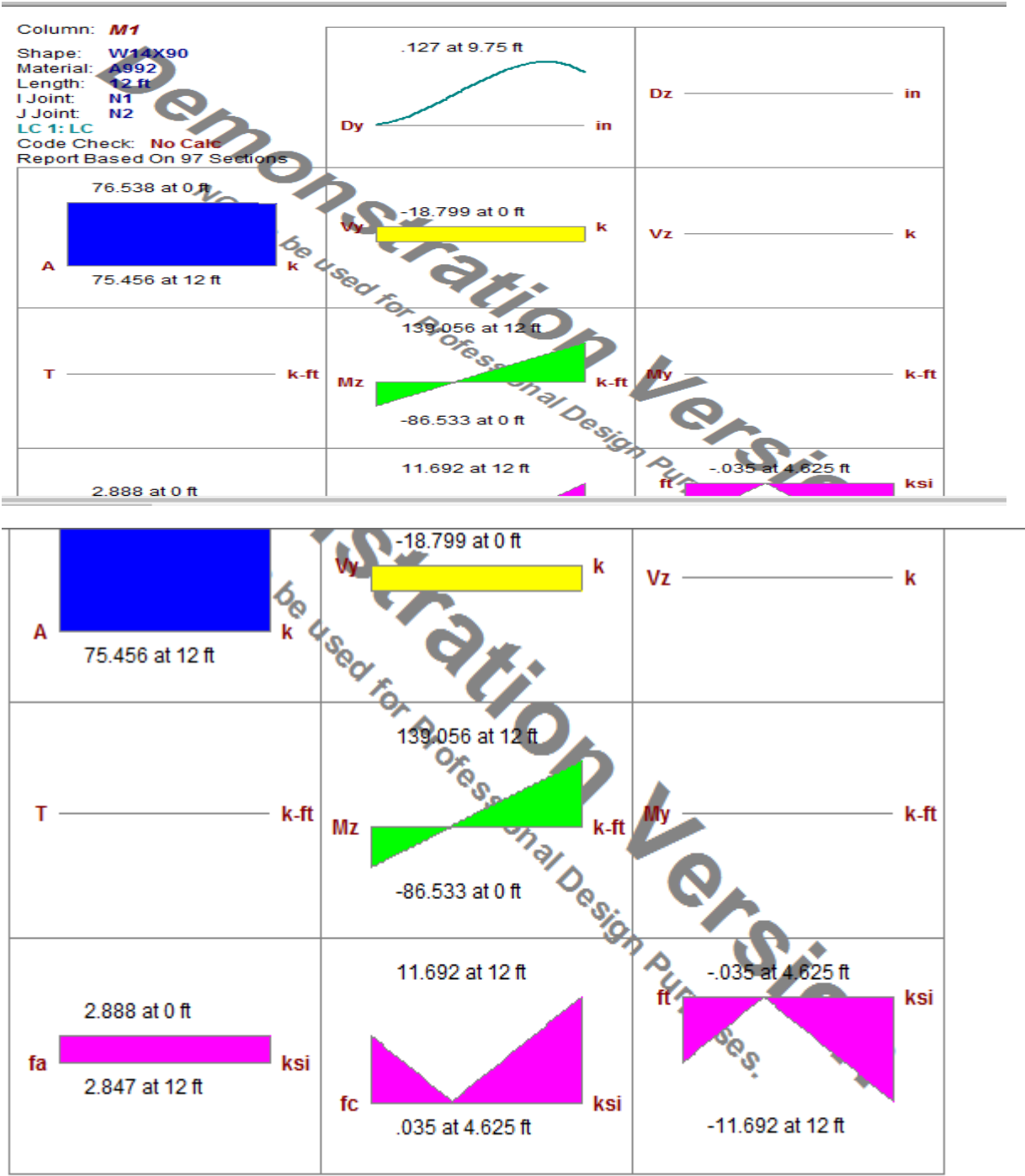
Create a stable frame structure in Risa 3D then compare the results to my handwritten calculations.

Results:

The following table and data shows the minimum and maximum shear and moment throughout the frame due to continuous loading being applied as stated in my lab 6 handout along with my handwritten calculations and Risa 3D output in order to create the model.

Figure 1: Frame created using Risa 3D with loading applied

Data: Data was collected using the details section of Risa 3D (shear and moment)



AISC 15th(360-16): ASD Code Check
Direct Analysis Method
- P-Delta analysis required for all AISC 360-16 Load Combinations -
Max Defl Ratio L/3092

Column: **M2**

Shape: **W14X90**

Material: **A992**

Length: **14 ft**

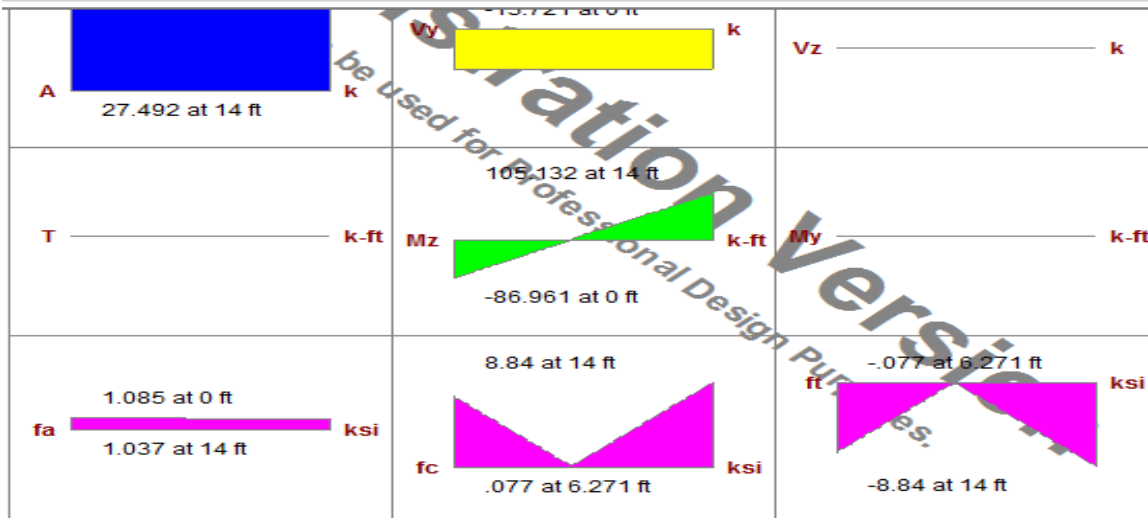
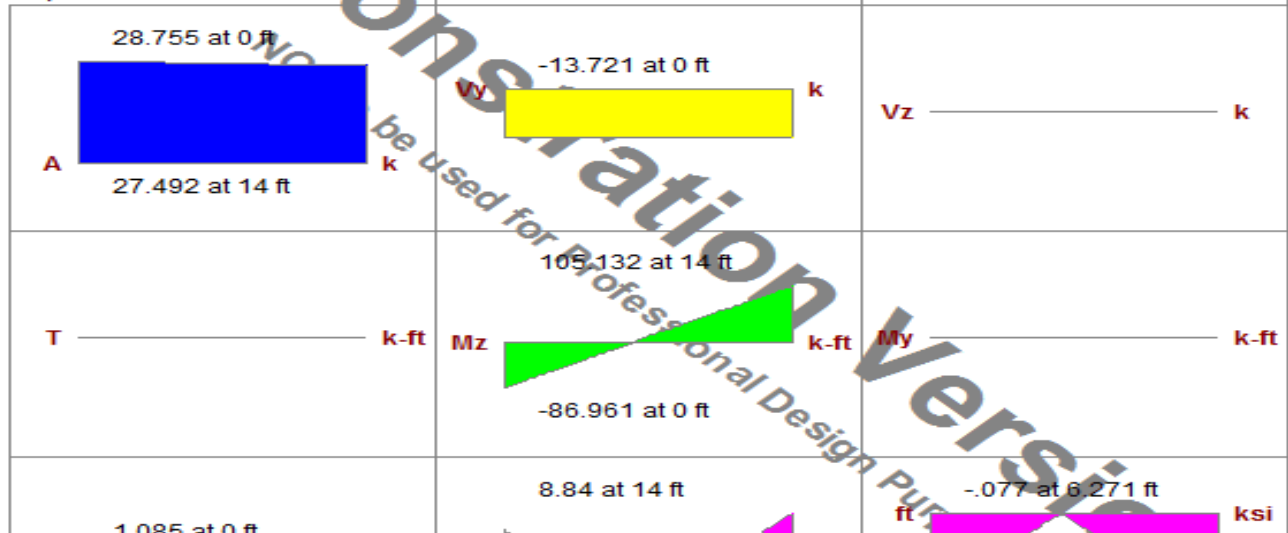
I Joint: **N2**

J Joint: **N3**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio **L/4887**

Column: **M3**

Shape: **W14X90**

Material: **A992**

Length: **12 ft**

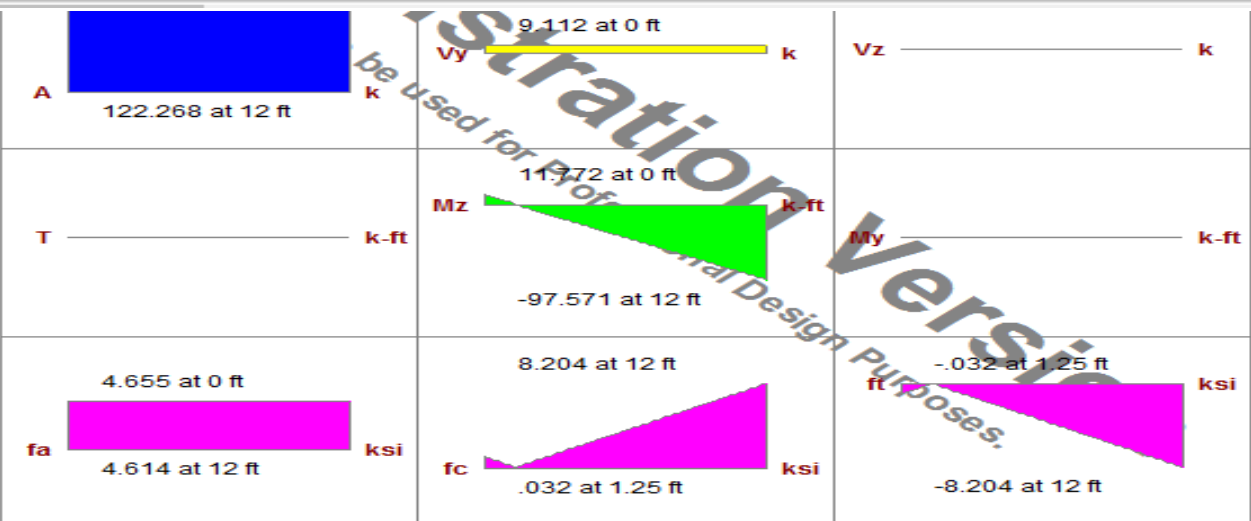
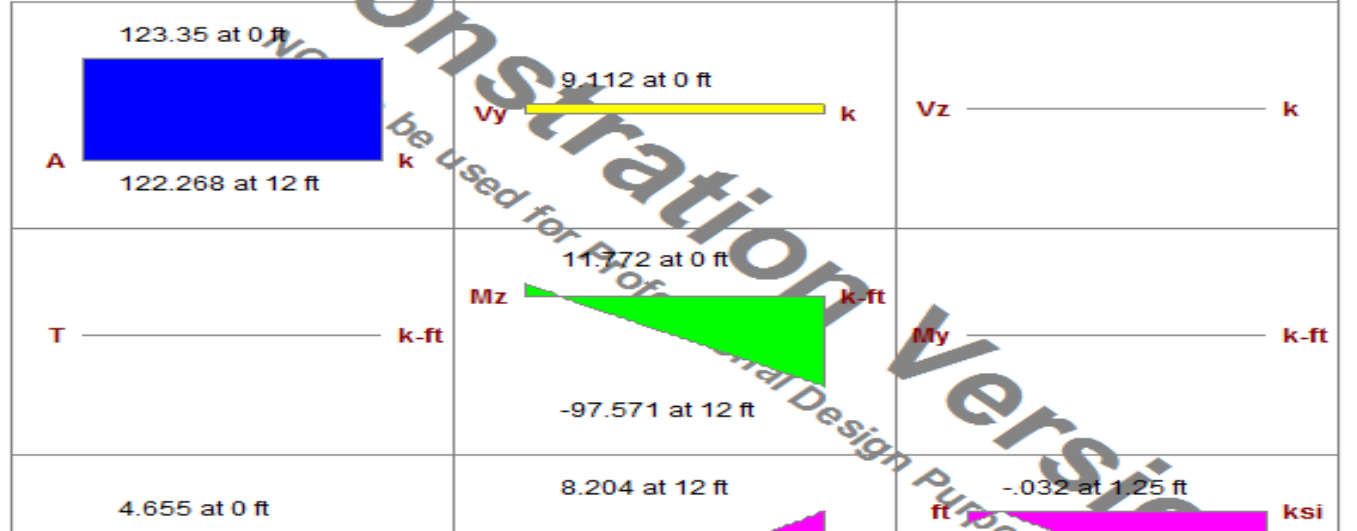
I Joint: **N4**

J Joint: **N5**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections

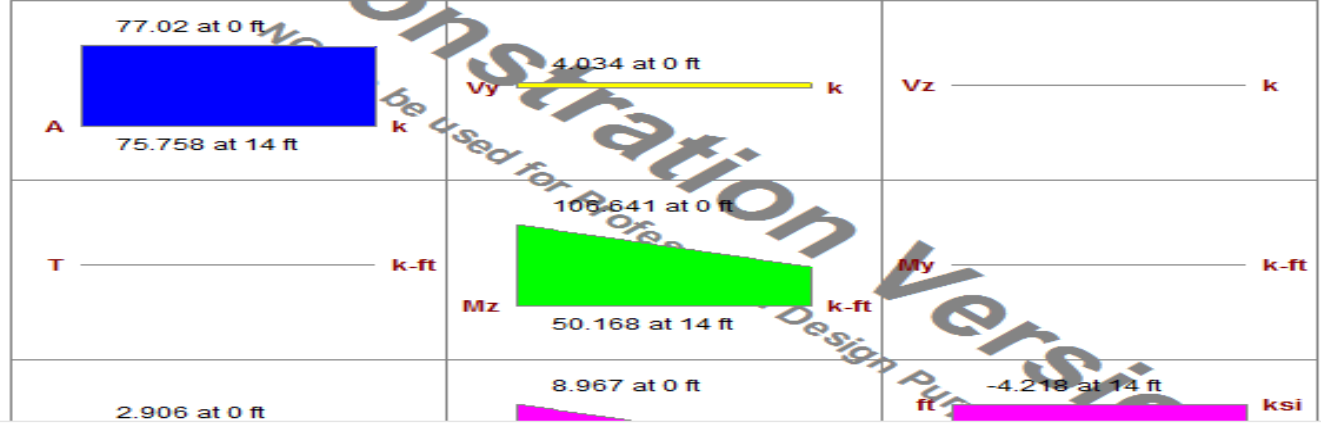


AISC 15th(360-16): ASD Code Check
Direct Analysis Method

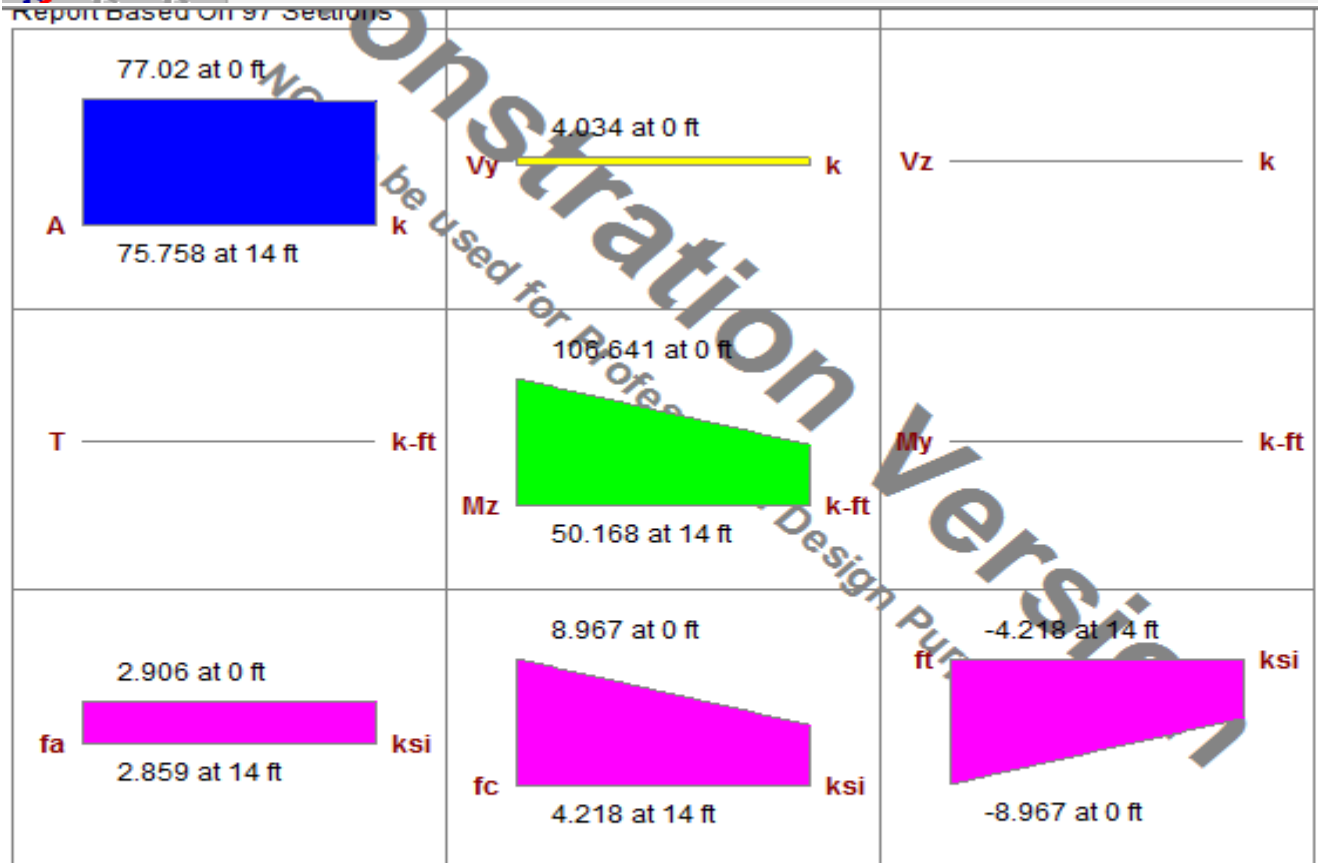
- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio **L/1304**

Column: **M4**
 Shape: **W14X90**
 Material: **A992**
 Length: **14 ft**
 I Joint: **N5**
 J Joint: **N6**
 LC 1: **LC**
 Code Check: **No Calc**
 Report Based On **97** Sections



Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check
 Direct Analysis Method

Column: **M5**

Shape: **W14X90**

Material: **A992**

Length: **12 ft**

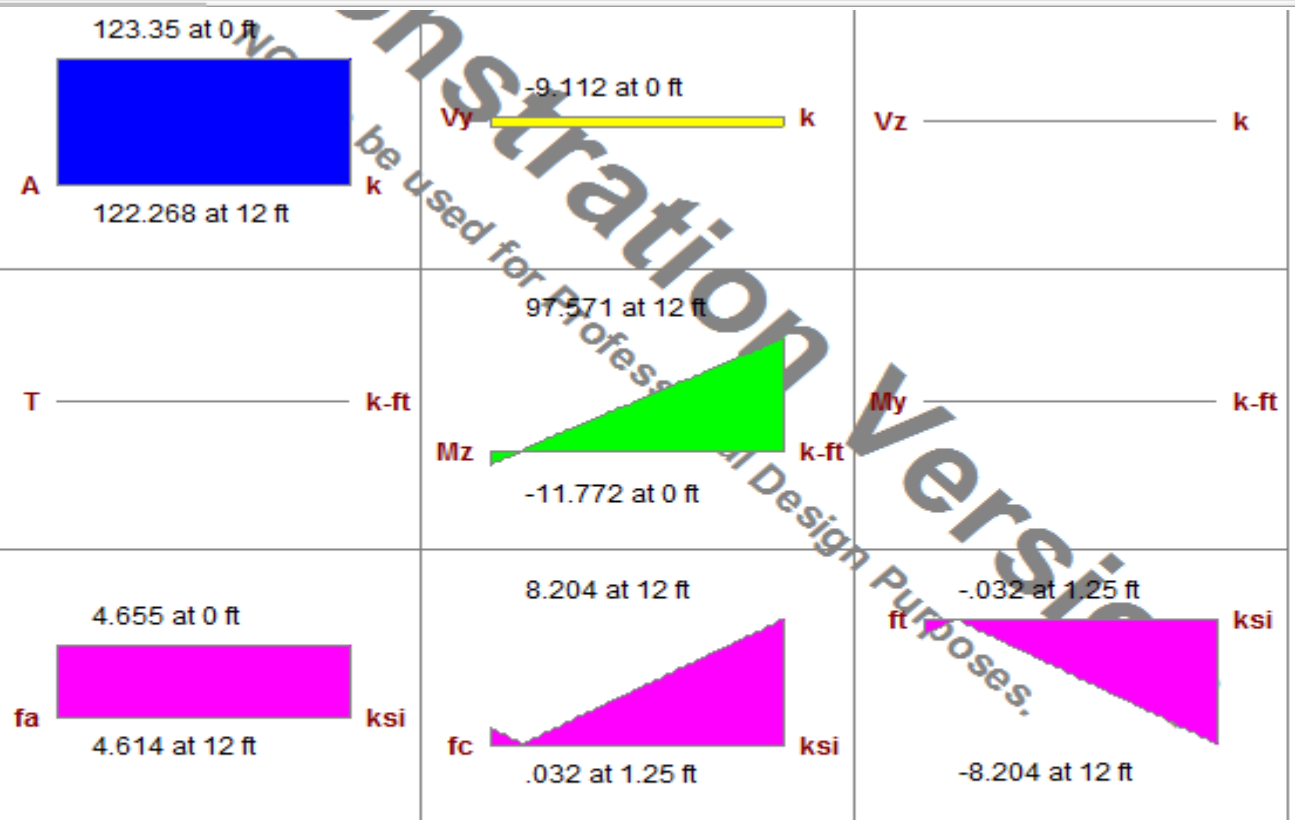
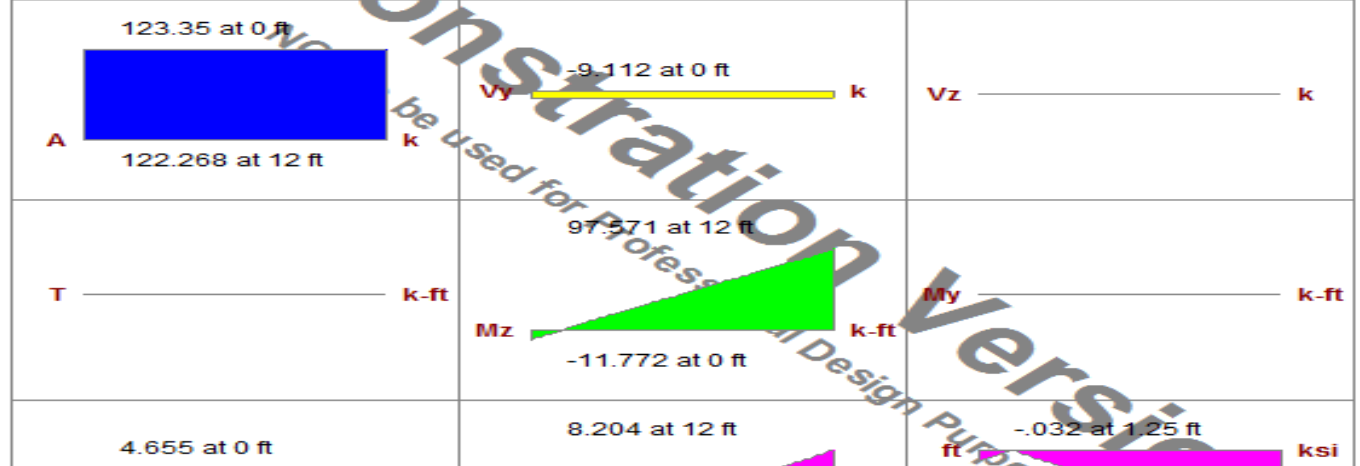
I Joint: **N7**

J Joint: **N8**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Column: **M6**

Shape: **W14X90**

Material: **A992**

Length: **14 ft**

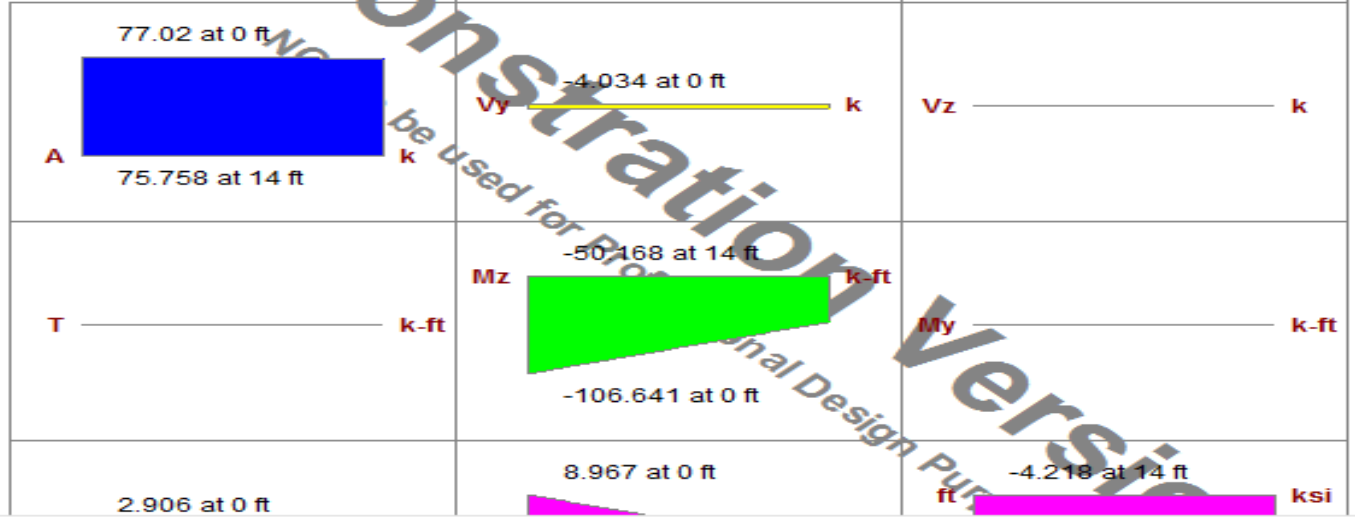
I Joint: **N8**

J Joint: **N9**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Column: **M7**

Shape: **W14X90**

Material: **A992**

Length: **12 ft**

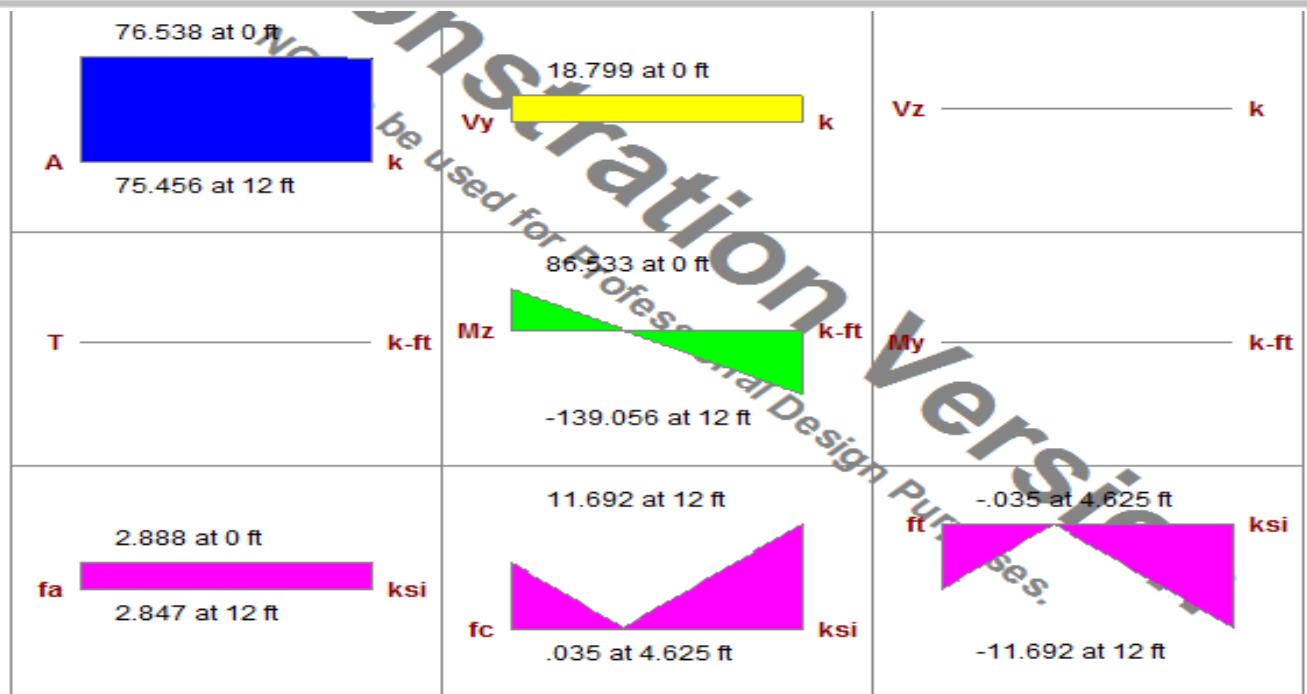
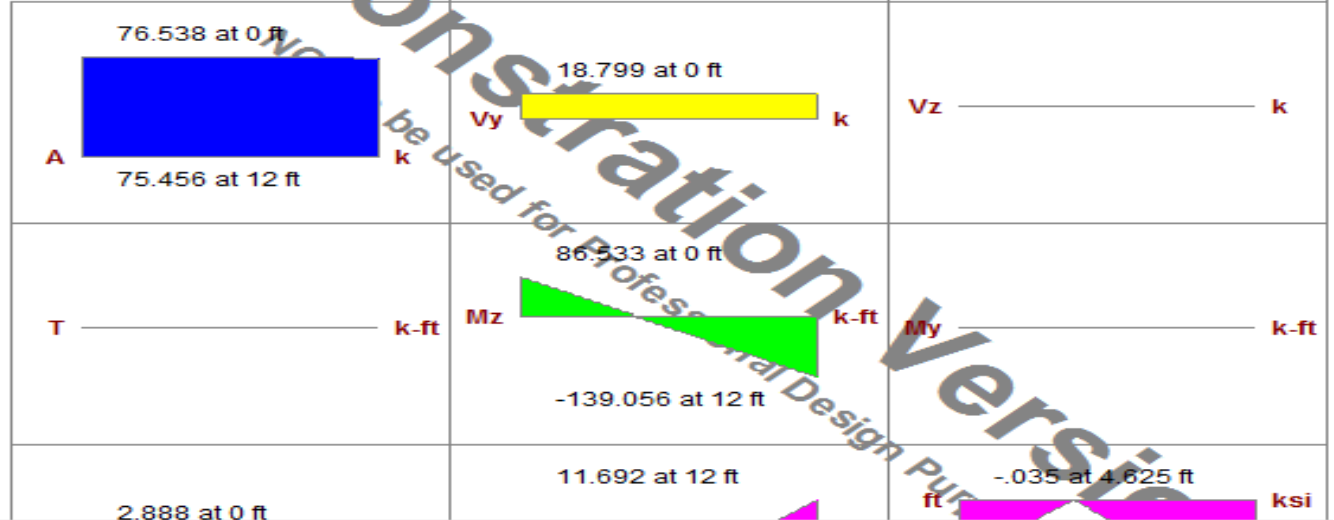
I Joint: **N10**

J Joint: **N11**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Column: **M8**

Shape: **W14X90**

Material: **A992**

Length: **14 ft**

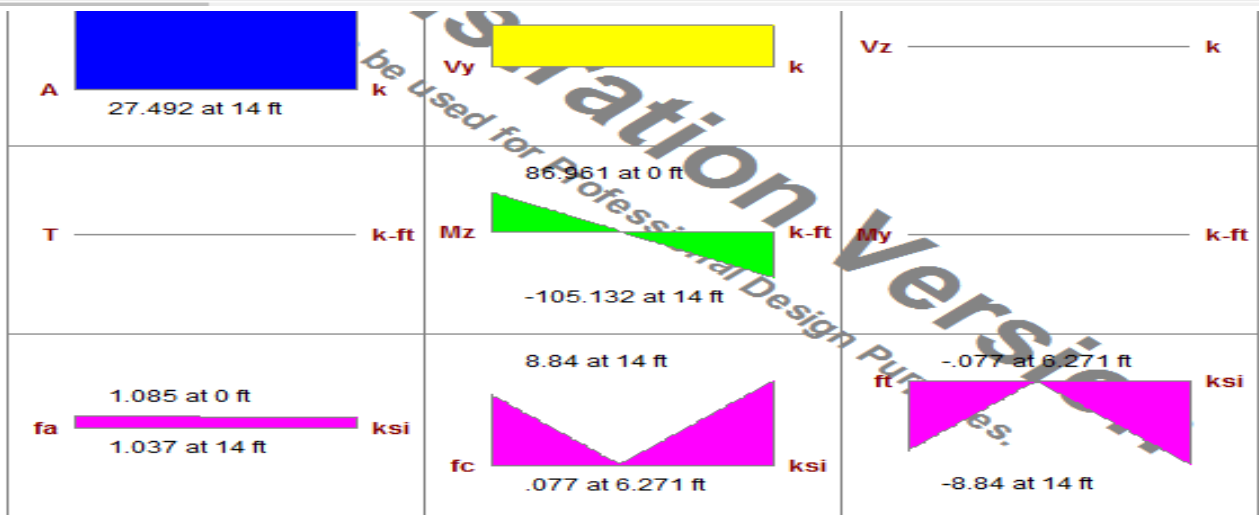
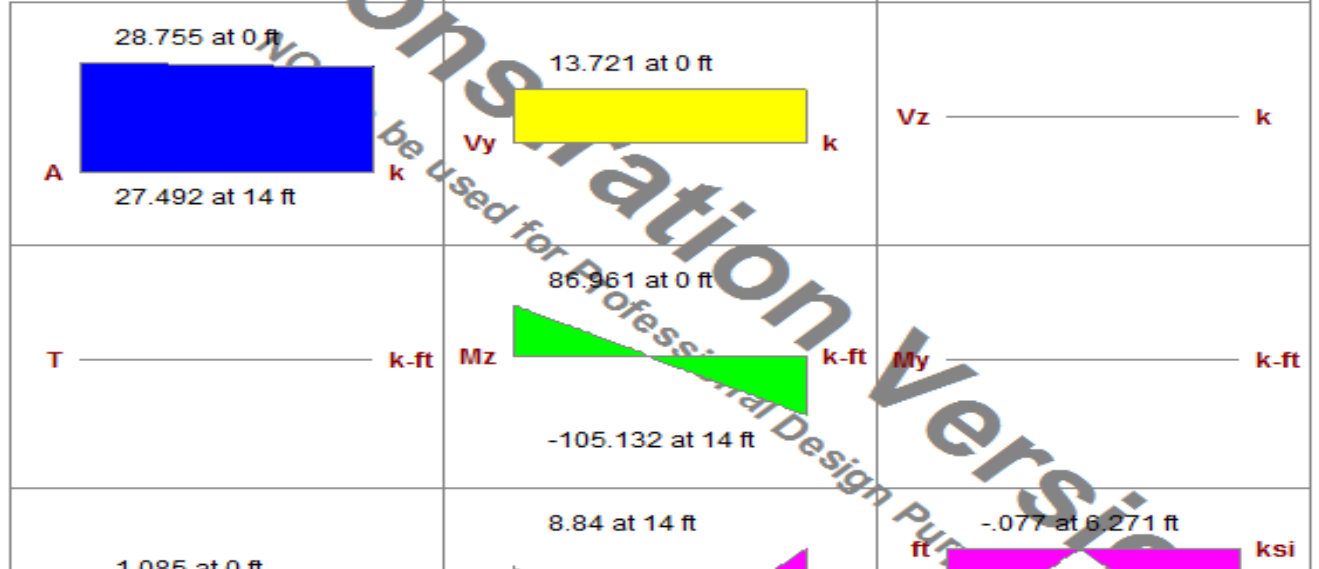
I Joint: **N11**

J Joint: **N12**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio **L/4887**

Beam: **M9**

Shape: **W12X65**

Material: **A992**

Length: **30 ft**

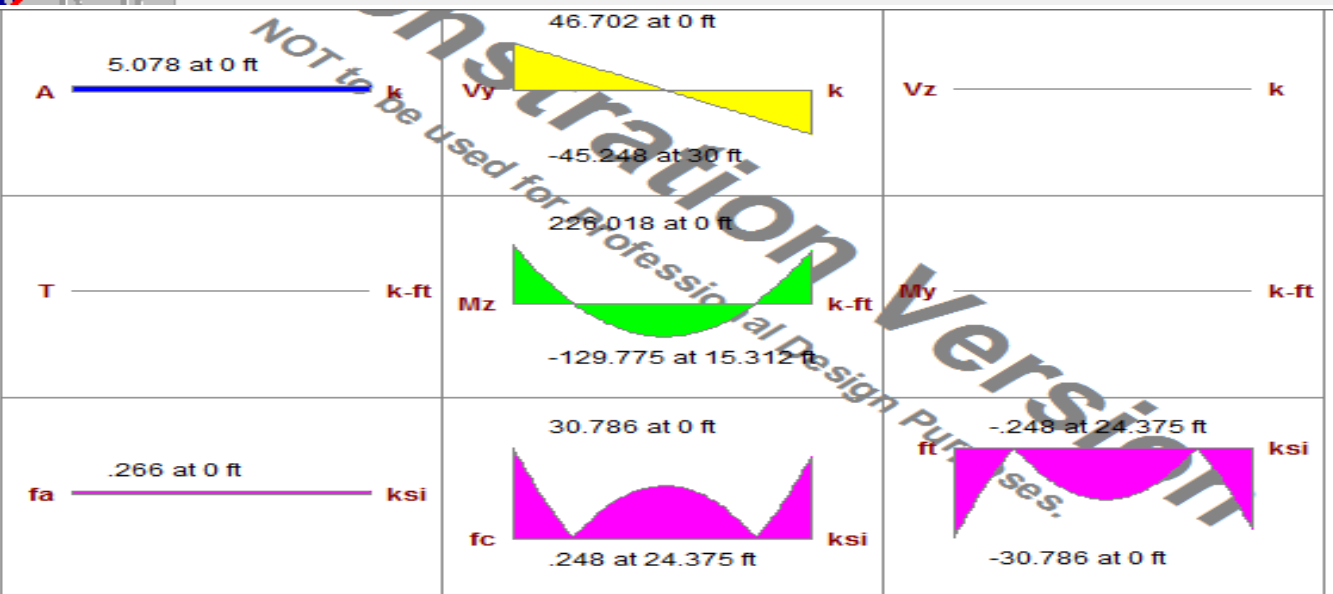
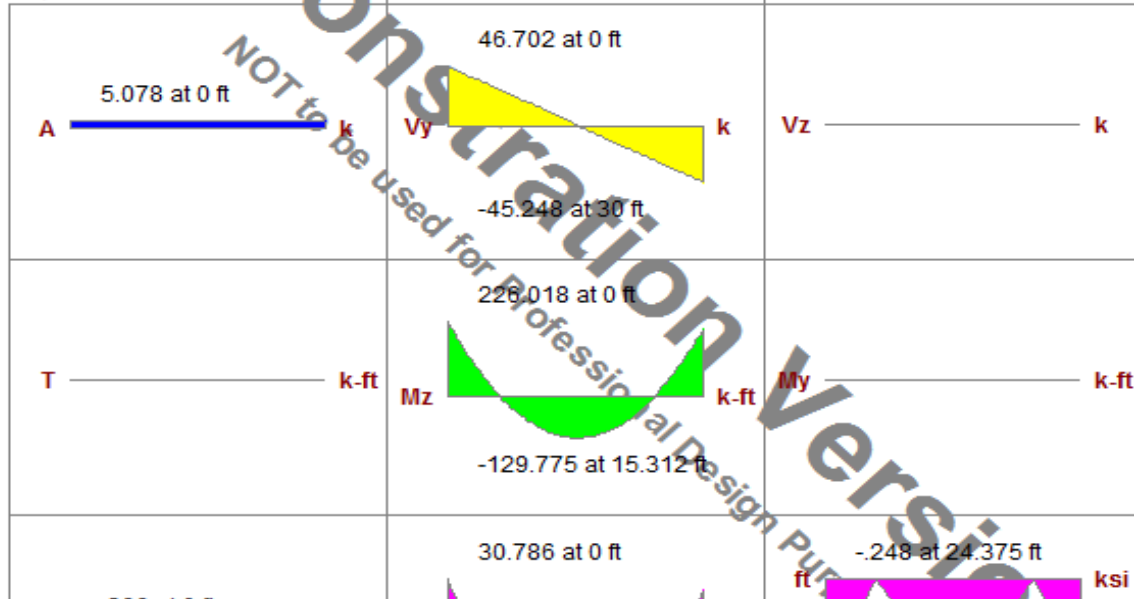
I Joint: **N2**

J Joint: **N5**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Beam: **M10**

Shape: **W12X65**

Material: **A992**

Length: **30 ft**

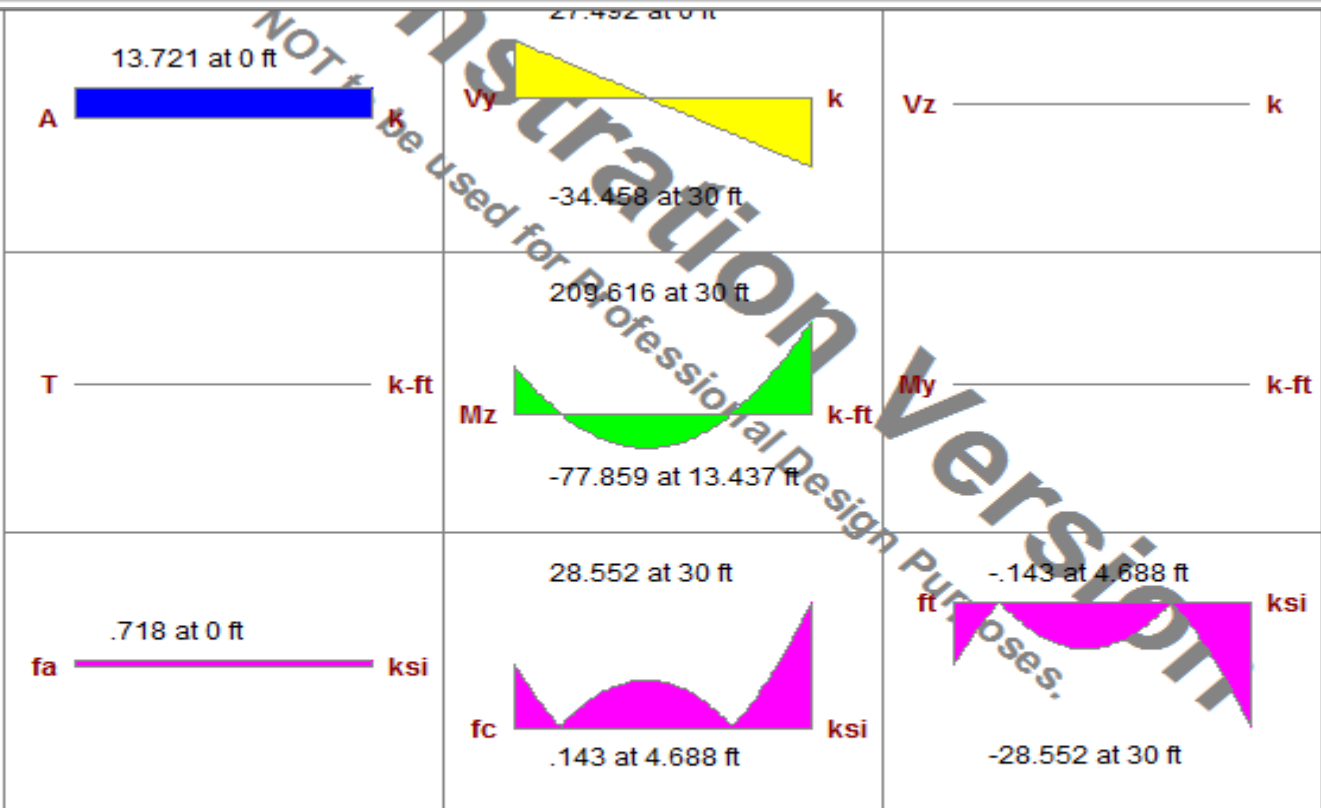
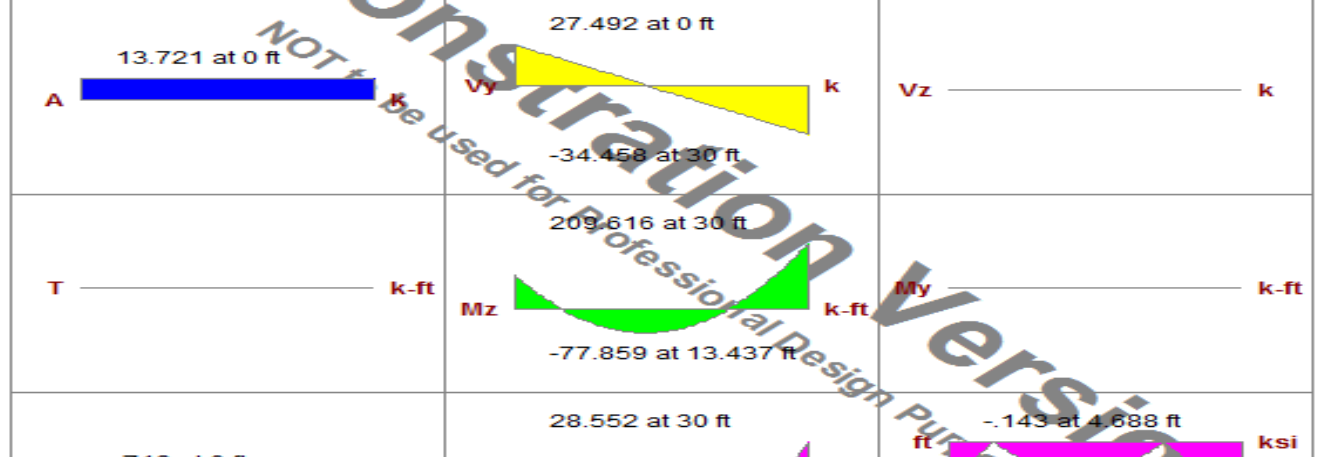
I Joint: **N3**

J Joint: **N6**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections

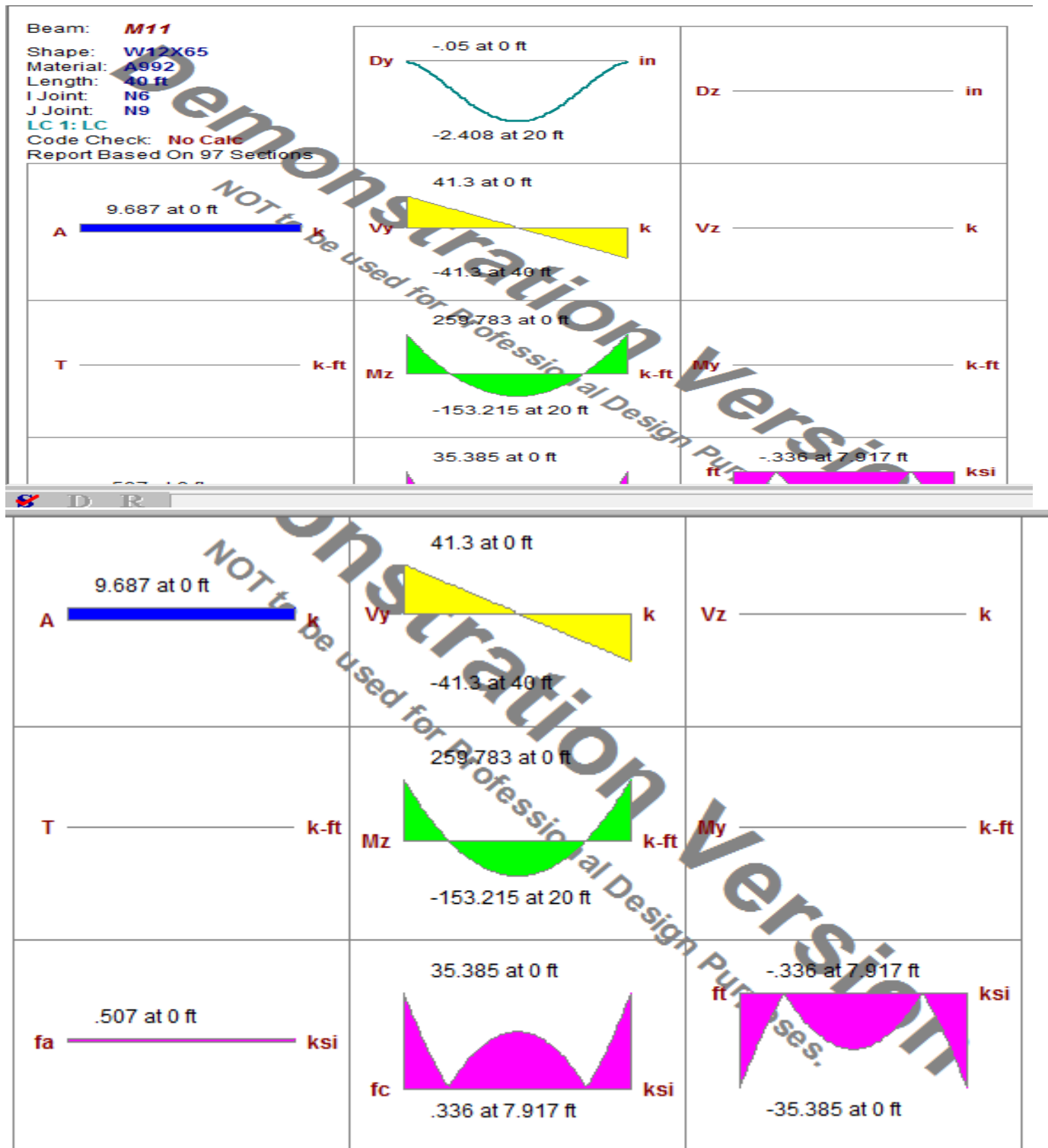


AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio: **1/615**



AISC 15th(360-16): ASD Code Check
 Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Beam: **M12**

Shape: **W12X65**

Material: **A992**

Length: **30 ft**

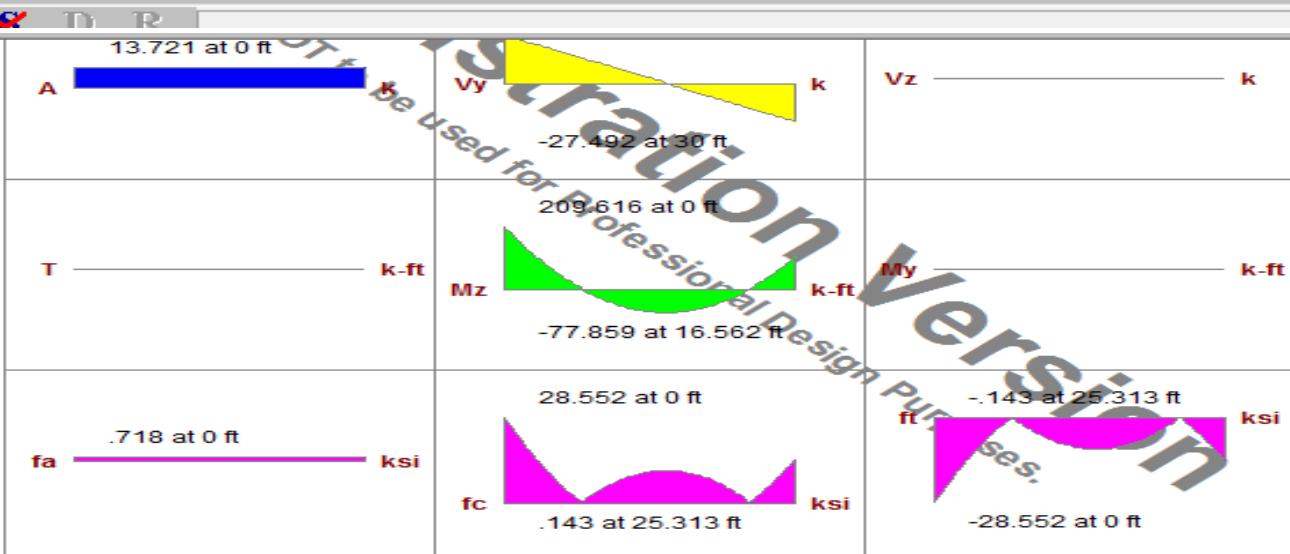
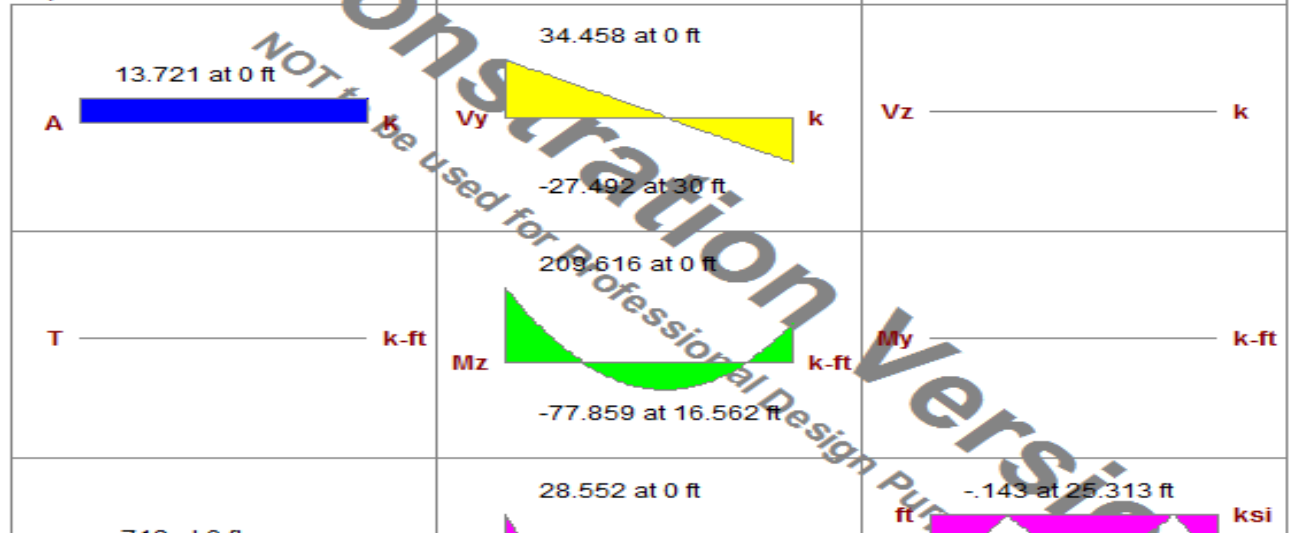
I Joint: **N9**

J Joint: **N12**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio **L/615**

Beam: **M13**

Shape: **W12X65**

Material: **A992**

Length: **30 ft**

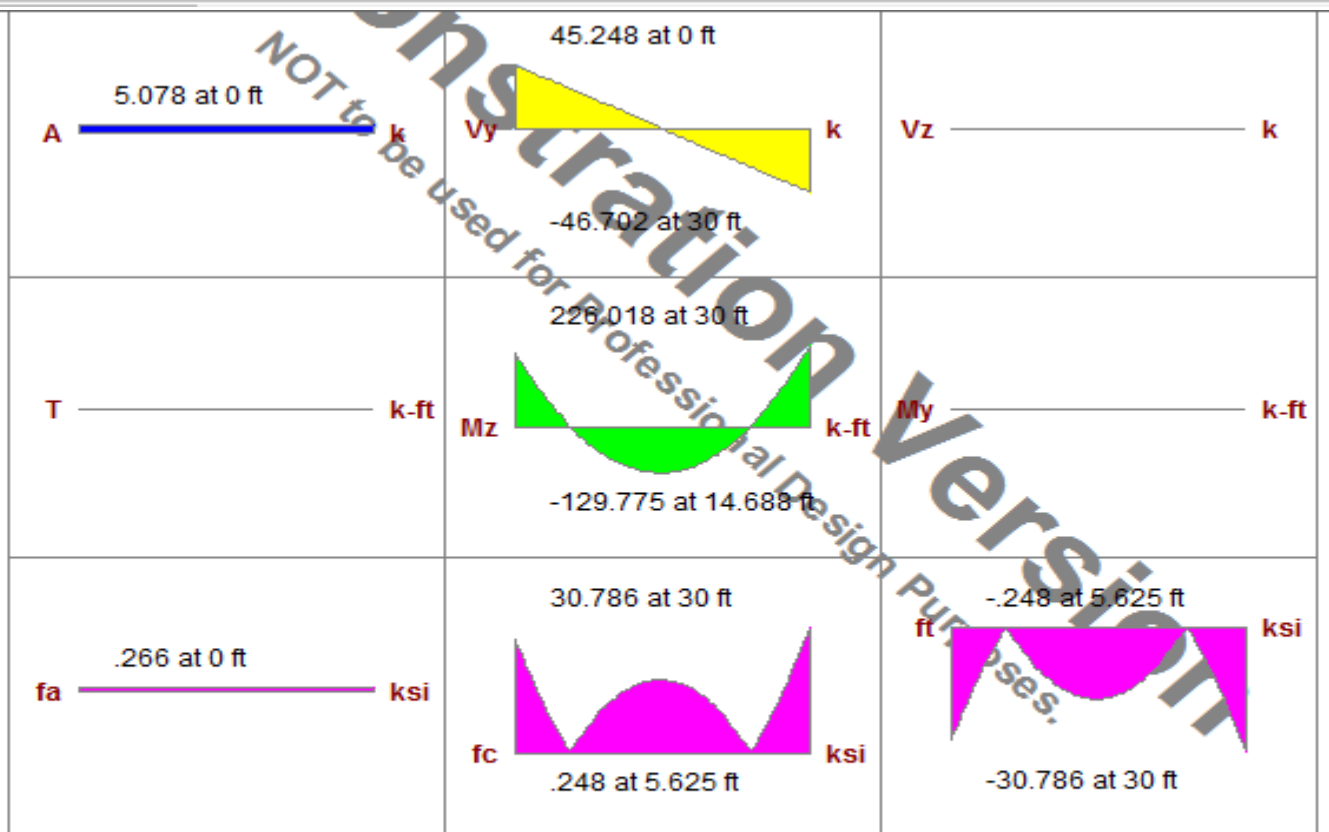
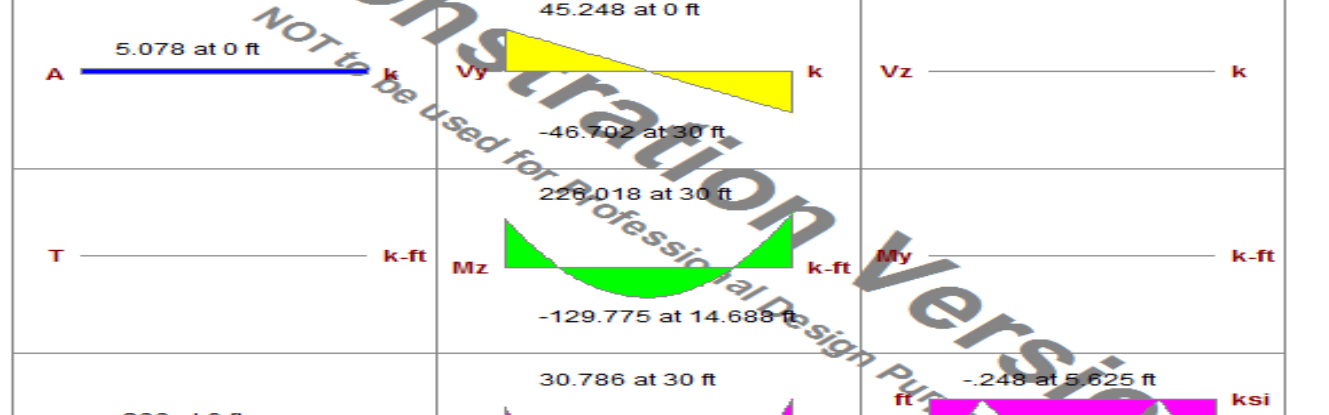
I Joint: **N8**

J Joint: **N11**

LC 1: **LC**

Code Check: **No Calc**

Report Based On 97 Sections



AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Conclusion:

In order to complete my model I used techniques from my previous lab reports along with the information given to me by my instructor regarding the exact kind of shape to use for my beams and columns. Once my frame was created I then applied my continuous loading on to each member as stated in my lab handout. After applying my joint coordinates, basic load cases, boundary conditions, and load cases I was able to run my results. (Results for my frame are located in the results section above.)

Using Risa 3D has given me information about each section of my truss in regards to my maximum and minimum shear and moment of that specific section. Shear and moment can be seen graphically in my results section under data, shear is shown using V_y while moment is shown using M . You can also understand whether you are looking at a beam or column by first looking in the description area of each picture where you will also find the column or beams shape. With this information you can determine that I constructed the frame using Risa 3D as it was intended to be and my handwritten calculations compared to Risa 3D are accurate.